

PROPOSAL

for

SEMESTER System Curriculum for B.Sc. (Ag.)

As Per:

***Vth DEANS' Committee
ICAR, New Delhi***

Semester-II

Sl. No.	Subject Name	Course Code	Course Title	Credit Hours
1	Agricultural Economics&Statistics	AES-121	Fundamentals of Agricultural Economics	2(2+0)
2	Agronomy	AGR-121	Farming System and Sustainable Agriculture	1(1+0)
3	Animal Husbandry & Dairy	AHD-121	Introductory Animal Husbandry	3(2+1)
4	Agricultural Extension	EXT-121	Fundamentals of Agricultural Extension Education	3(2+1)
5	Entomology	ENT-121	Fundamental of Entomology-I	2(1+1)
6	Genetics & Plant Breeding	GPB-121	Fundamental of Crop Physiology	2(1+1)
7	Horticulture	HOR-121	Production Technology for Fruit and Plantation Crops	2(1+1)
8	Plant Pathology	PPA-121	Fundamental of Plant Pathology	4(3+1)
9	Soil Science & Agricultural Chemistry	SSC-121	Fundamentals of Plant Biochemistry	2(1+1)
10	Soil Conservation	SCW-121	Soil and Water Conservation Engineering	2(1+1)
			Total	23

* - Remedial

** - NonGradial

Semester II

Course Title: Fundamentals of Agricultural Economics

Course Code: AES-121

CreditHours:: 2 (2+0)

UNIT I: Economics: Meaning, scope and subject matter, definitions, activities, approaches to economic analysis; micro and macro economics, positive and normative analysis. Nature of economic theory; rationality assumption, concept of equilibrium, economic laws as generalization of human behaviour.

UNIT II: Basic concepts: Goods and services, desire, want, demand, utility, cost and price, wealth, capital, income and welfare.

UNIT III: Agricultural economics : meaning, definition, characteristics of agriculture, importance and its role in economic development. Agricultural planning and development in the country.

UNIT IV: Demand: meaning, law of demand, schedule and demand curve, determinants, utility theory; law of diminishing marginal utility, equi-marginal utility principle. Consumer's equilibrium and derivation of demand curve, concept of consumer surplus.

UNIT V: Elasticity of demand: concept and measurement of price elasticity, income elasticity and cross elasticity.

UNIT VI: Production: Process, creation of utility, factors of production, input output relationship.

UNIT VII: Laws of returns: Law of variable proportions and law of returns to scale. Cost: concepts, short run and long run cost curves. Supply law of supply, schedule, supply curve, determinants of supply, and elasticity of supply.

UNIT VIII: Market structure: meaning and types of market, basic features of perfectly competitive and imperfect markets. Price determination under perfect competition; short run and long run equilibrium of firm and industry, shut down and break even points.

UNIT IX: Distribution theory: meaning, factor market and pricing of factors of production. Concepts of rent, wage, interest and profit.

UNIT X: National income: Meaning and importance, circular flow, concepts of national income accounting and approaches to measurement, difficulties in measurement.

UNIT XI: Population: Importance, Malthusian and optimum population theories, natural and socio-economic determinants, current policies and programmes on population control.

UNIT XII: Money: Barter system of exchange and its problems, evolution, meaning and functions of money, classification of money, supply, general price index, inflation and deflation. Public revenue and public expenditure.

UNIT XIII: Tax: meaning, direct and indirect taxes, agricultural taxation, VAT, GST.

UNIT XIV: Economic systems: Concepts of economy and its functions, important features of capitalistic, socialistic and mixed economies, elements of economic planning.

Course Title: Farming System and Sustainable Agriculture

Course Code: AGR-121

CreditHours:: 1(1+0)

Theory

Farming System-scope, importance, and concept, Types and systems of farming system and factors affecting types of farming, Farming system components and their maintenance, Cropping system and pattern, multiple cropping system, Efficient cropping system and their evaluation, Allied enterprises and their importance, Tools for determining production and efficiencies in cropping and farming system; Sustainable agriculture-problems and its impact on agriculture, indicators of sustainability, adaptation and mitigation, conservation agriculture strategies in agriculture, HEIA, LEIA and LEISA and its techniques for sustainability, Integrated farming system-historical background, objectives and characteristics, components of IFS and its advantages, Site specific development of IFS model for different agro-climatic zones, resource use efficiency and optimization techniques, Resource cycling and flow of energy in different farming system, farming system and environment, Visit of IFS model in different agro-climatic zones of nearby states University/ institutes and farmers field.

Course Title: Introductory Animal Husbandry

Course Code:AHD-121

CreditHours:: 3(2+1)

Theory

Importance of livestock in agriculture and economy. Dairying under specialized and mixed farming. Livestock and milk production statistics. Dairy Cattle and Buffaloes management: cattle and buffaloes breeds, breeding methods and systems, care and management of pregnant and mulching animals, raising of calves, management of heifer and bulls, maintenance of livestock records, milking methods and principles, clean milk production, feed and feeding, Pig, Management: importance, important breeds, raising of piglet up to age of slaughter, general aspect of breeding, care of sow and boar. Sheep and goat: importance, important breeds, raising of kids and lambs, breeding, feeding of goat and sheep. Health Management: Common animal diseases of cattle, buffalo, goat, sheep and swine viz. anthrax, BQ, HS, Brucellosis, Mastitis, swine fever and enterotoxaemia, vaccination schedule.

Practical

Judging of cattle, sheep and wool. Detection of heat and pregnancy, computation of ration for different purposes of different animals. Milking methods, record keeping and use of quadrille in swine.

Course Title: Fundamentals of Agricultural Extension Education

Course Code: EXT-121

CreditHours:: 3 (2+1)

Theory

Education: Meaning, definition & Types; Extension Education- meaning, definition, scope and process; objectives and principles of Extension Education; Extension Programme planning- Meaning, Process, Principles and Steps in Programme Development. Extension systems in India: extension efforts in pre-independence era (Sriniketan, Marthandam, Firka Development Scheme, Gurgaon Experiment, etc.) and post-independence era (Etawah Pilot Project, Nilokheri Experiment, etc.); various extension/ agriculture development programmes launched by ICAR/ Govt. of India (IADP, IAAP, HYVP, KVK, IVLP, ORP, ND,NATP, NAIP, etc.). New trends in agriculture extension: privatization extension, cyber extension/ e-extension, market-led extension, farmer-led extension, expert systems, etc.

Rural Development: concept, meaning, definition; various rural development programmes launched by Govt. of India. Community Dev.-meaning, definition, concept & principles, Philosophy of C.D. Rural Leadership: concept and definition, types of leaders in rural context; extension administration: meaning and concept, principles and functions. Monitoring and evaluation: concept and definition, monitoring and evaluation of extension programmes; transfer of technology: concept and models, capacity building of extension personnel; extension teaching methods: meaning, classification, individual, group and mass contact methods, ICT Applications in TOT (New and Social Media), media mix strategies; communication: meaning and definition; Principles and Functions of Communication, models and barriers to communication. Agriculture journalism; diffusion and adoption of innovation: concept and meaning, process and stages of adoption, adopter categories.

Practical

To get acquainted with university extension system. Group discussion- exercise; handling and use of audio visual equipments and digital camera and LCD projector; preparation and use of AV aids, preparation of extension literature – leaflet, booklet, folder, pamphlet news stories and success stories; Presentation skills exercise; micro teaching exercise; A visit to village to understand the problems being encountered by the villagers/ farmers; to study organization and functioning of DRDA and other development departments at district level; visit to NGO and learning from their experience in rural development; understanding PRA techniques and their application in village development planning; exposure to mass media: visit to community radio and television studio for understanding the process of programme production; script writing, writing for print and electronic media, developing script for radio and television.

Course Title: Fundamentals of Entomology-I

Course Code: ENT-121

CreditHours:: 2(1+1)

Theory

History of Entomology in India. Major points related to dominance of Insecta in Animal kingdom. Classification of phylum Arthropoda upto classes. Relationship of class Insecta with other classes of Arthropoda.

Morphology:

Structure and functions of insect cuticle and molting. Body segmentation. Structure of Head, thorax and abdomen. Structure and modifications of insect antennae, mouth parts, legs, Wing venation, modifications and wing coupling apparatus. Metamorphosis and diapause in insects. Types of larvae and pupae.

Structure and functions of digestive, excretory, respiratory, secretory and reproductive system, in insects. Types of reproduction in insects. Major sensory organs like simple and compound eyes, chemoreceptor.

Systematics:

Taxonomy –importance, history and development and binomial nomenclature. Definitions of Biotype, Sub-species, Species, Genus, Family and Order. Classification of class Insecta upto Orders, basic groups of present day insects with special emphasis to orders and families of Agricultural importance like

Orthoptera: Acrididae, Tettigonidae, Gryllidae, Gryllotalpidae;

Dictyoptera: Mantidae, Blattidae;

Odonata;

Isoptera: Termitidae;

Thysanoptera: Thripidae;

Hemiptera: Pentatomidae, Coreidae, Cimicidae, Pyrrhocoridae, Lygaeidae, Cicadellidae, Delphacidae, Aphididae, Coccidae, Lophophidae, Aleurodidae, Pseudococcidae; Neuroptera: Chrysopidae;

Lepidoptera: Pieridae, Papilionidae, Noctuidae, Sphingidae, Pyralidae, Gelechiidae, Arctiidae, Saturnidae, Bombycidae;

Coleoptera: Coccinellidae, Chrysomelidae, Cerambycidae, Curculionidae, Bruchidae, Scarabaeidae;

Hymenoptera: Tenthredinidae, Apidae, Trichogrammatidae, Ichneumonidae, Braconidae, Chalcididae;

Diptera: Cecidomyiidae, Tachinidae, Agromyziidae, Culicidae, Muscidae, Tephritidae.

Practical

Methods of collection and preservation of insects including immature stages; External features of Grasshopper/Blister beetle; Types of insect antennae, mouthparts and legs; Wing venation, types of wings and wing coupling apparatus. Types of insect larvae and pupae; Dissection of digestive system in insects (Grasshopper); Dissection of male and female reproductive systems in insects (Grasshopper);

Study of characters of orders Orthoptera, Dictyoptera, Odonata, Isoptera, Thysanoptera, Hemiptera, Lepidoptera, Neuroptera, Coleoptera, Hymenoptera, Diptera and their families of agricultural importance.

Course Title: Fundamentals of Crop Physiology

Course Code: GPB-121

Credit Hours: 2(1+1)

Theory

Introduction to crop physiology and its importance in Agriculture; Plant cell: an Overview; Diffusion and osmosis; Absorption of water, transpiration and Stomatal Physiology; Mineral nutrition of Plants: Functions and deficiency symptoms of nutrients, nutrient uptake mechanisms; Photosynthesis: Light and Dark reactions, C3, C4 and CAM plants; Respiration: Glycolysis, TCA cycle and electron transport chain; Fat Metabolism. Plant growth regulators: Physiological roles and agricultural uses, Physiological aspects of growth and development of major crops: Growth analysis, Role of Physiological growth parameters in crop productivity.

Practical

Study of plant cells, structure and distribution of stomata, imbibitions, osmosis, plasmolysis, measurement of root pressure, rate of transpiration, Separation of photosynthetic pigments through paper chromatography, Rate of transpiration, photosynthesis, respiration, tissue test for mineral nutrients, estimation of relative water content, Measurement of photosynthetic.

Course Title: Production Technology for Fruit and Plantation Crops

Course Code: HOR-121

Credit Hours: 2(1+1)

Theory

Importance and scope of fruit and plantation crop industry in India; Importance of rootstocks; Production technologies for the cultivation of major fruits-mango, banana, citrus, grape, guava, litchi, papaya, sapota, apple, pear, peach, walnut, almond and; minor fruits- date, ber, pineapple, pomegranate, jackfruit, strawberry, plantation crops-coconut, arecanut, cashew, tea, coffee & rubber.

Practical

Seed propagation. Scarification and stratification of seeds. Propagation methods for fruit and plantation crops. Description and identification of fruit. Preparation of plant bio regulators and

their uses, Important pests, diseases and physiological disorders of above fruit and plantation crops, Visit to commercial orchards.

Course Title: Fundamentals of Plant Pathology

Course Code:PPA-121

Credit Hours: 4(3+1)

Theory

Introduction: Importance of plant diseases, scope and objectives of Plant Pathology. History of Plant Pathology with special reference to Indian work. Terms and concepts in Plant Pathology. Causes / factors affecting disease development: disease triangle and tetrahedron and classification of plant diseases. Important plant pathogenic organisms, different groups: fungi, bacteria, fastidious vesicular bacteria, phytoplasmas, spiroplasmas, viruses, viroids, algae, protozoa, phanerogamic parasites and nematodes with examples of diseases caused by them. Diseases and symptoms due to abiotic causes.

Fungi: general characters, definition of fungus, somatic structures, types of fungal thalli, fungal tissues, modifications of thallus, reproduction (asexual and sexual). Nomenclature, Binomial system of nomenclature, rules of nomenclature, classification of fungi. Key to divisions, subdivisions, orders and classes.

Life cycles of *Pythium*, *Phytophthora*, *Albugo*, *Peronospora*, *Sclerospora*, *Ustilago*, *Sphacelotheca*, *Tolyposporium*, *Mucor*, *Rhizopus*, *Erysiphe*, *Aspergillus*, *Penicillium*, *Puccinia*, *Melampsora*, *Claviceps*, *Alternaria*, *Collectotrichum*, *Cercospora*, *Helminthosporium*, *Pyricularia*, *Rhizoctonia*. and *Fusarium*.

Bacteria and mollicutes: general morphological characters. Basic methods of classification and reproduction.

Viruses: nature, structure, replication and transmission. Study of phanerogamic plant parasites.

Nematodes: General morphology and reproduction, classification, symptoms and nature of damage caused by plant nematodes (*Heterodera*, *Meloidogyne*, *Anguina*).

Liberation / dispersal and survival of plant pathogens. Role of enzymes, toxins and growth regulators in disease development. Defense mechanism in plants. Elementary knowledge of pathogenesis, parasitism, symbiosis and antibiosis. Epidemiology: Factors affecting disease development. Principles and methods of plant disease management. Nature, chemical combination, classification, mode of action and formulations of fungicides and antibiotics.

Practical

Acquaintance with various laboratory equipments and microscopy. Collection and preservation of disease specimen. Preparation of media, isolation and Koch's postulates. General study of different structures of fungi. Study of symptoms of various plant diseases. Study of representative fungal genera. Staining and identification of plant pathogenic bacteria. Transmission of plant viruses. Study of phanerogamic plant parasites. Study of morphological features and identification of plant parasitic nematodes. Sampling and extraction of nematodes from soil and plant material, preparation of nematode mounting. Study of fungicides and their formulations. Methods of pesticide application and their safe use. Calculation of fungicide sprays concentrations.

Course Title: Fundamentals of Plant Biochemistry

Course Code: SSC-121

Credit Hours: 2(1+1)

Theory

Importance of Biochemistry. Properties of Water, pH and Buffer. Carbohydrate: Importance and classification. Structures of Monosaccharides, Reducing and oxidizing properties of Monosaccharides, Mutarotation; Structure of Disaccharides and Poly saccharides. Lipid: Importance and classification; Structures and properties of fatty acids; storage lipids and membrane lipids. Proteins: Importance of proteins and classification; Structures, titration and zwitterions nature of amino acids; Structural organization of proteins. Enzymes: General properties; Classification; Mechanism of action; Michaelis&Menten and Line Weaver Burk equation & plots; Introduction to allosteric enzymes. Nucleic acids: Importance and classification; Structure of Nucleotides, A, B & Z DNA; RNA: Types and Secondary & Tertiary structure. Metabolism of carbohydrates: Glycolysis, TCA cycle, Glyoxylate cycle, Electron transport chain. Metabolism of lipids: Beta oxidation, Biosynthesis of fatty acids.

Practical

Preparation of solution, pH & buffers, Qualitative tests of carbohydrates and amino acids. Quantitative estimation of glucose/ proteins. Titration methods for estimation of amino acids/lipids, Effect of pH, temperature and substrate concentration on enzyme action, Paper chromatography/ TLC demonstration for separation of amino acids/ Monosaccharides.

Course Title: Soil and Water Conservation Engineering

Course Code: SCW-121

Credit Hours::2 (1+1)

1. **Soil and Water Conservation:** Meaning and definition, aims, Causes of soil erosion, Definition and agents of soil erosion, Forms of water erosion.
2. **Gully:** classification and control measures.
3. **Soil loss:** Soil loss estimation by universal Loss Soil Equation and Soil loss measurement techniques.
4. **Control measures of soil erosion:** Agronomical and mechanical. Introduction to contouring, strip cropping.
5. **Bunding:** Contour bund and Graded bund
6. **Terracing:** Bench terracing.
7. Grassed water ways and their design.
8. Water harvesting and its techniques.
9. **Wind erosion:** Mechanics of wind erosion, types of soil movement, Principles of wind erosion control and its control measures.

10. **Run-off**: Meaning and definition, type, factor affecting, estimation and measurement of run-off, Time of Concentration (Tc) and its computation.

Practical

1. General status of soil conservation in India.
2. Calculation of erosion index.
3. Estimation of soil loss.
4. Measurement of soil loss.
5. Preparation of contour maps.
6. Design of grassed water ways.
7. Design of contour bunds.
8. Design of graded bunds.
9. Design of bench terracing system.
10. Problem on wind erosion.
11. Computation of run-off by **Rational** method
12. Computation of Tc